

## **SOFT2412/COMP9412 Assignment**

### **Agile Software Development Practices**

**Unit of Study:** SOFT2412/COMP9412

**Sprint Number:** Sprint 1

**Tutorial time:** Friday 2pm-4pm

**Tutor name:** Amyra Meidiana

#### **DECLARATION**

We, the undersigned, declare that the work contained in this assignment/project is our own work and has not been copied from other sources or been previously submitted for award or assessment. We understand that failure to comply with the Student Plagiarism: Coursework Policy and Procedure can lead to severe penalties as outlined under Chapter 8 of the University of Sydney By-Law 1999 (as amended). These penalties may be imposed in cases where any significant portion of my submitted work has been copied without proper acknowledgement from other sources, including published works, the internet, existing programs, the work of other students, or work previously submitted for other awards or assessments. We realise that we may be asked to identify those portions of the work contributed by each of us and required to demonstrate our individual knowledge of the relevant material by answering oral questions or by undertaking supplementary work, either written or in the laboratory, in order to arrive at the final assessment mark.

| <b>Project Team Members</b> |                   |                  |
|-----------------------------|-------------------|------------------|
| <b>Student Name</b>         | <b>Student ID</b> | <b>Signature</b> |
| Zhiwei Cheng                | 540249451         | Zhiwei Cheng     |
| Xuejian Fang                | 530313995         | Xuejian Fang     |
| Boning Zhao                 | 530027564         | Boning Zhao      |
| Xuanhao Yu                  | 530349673         | Xuanhao Yu       |

# 1. Project Overview

## 1.1 Project Goals and Objectives

### Project goal

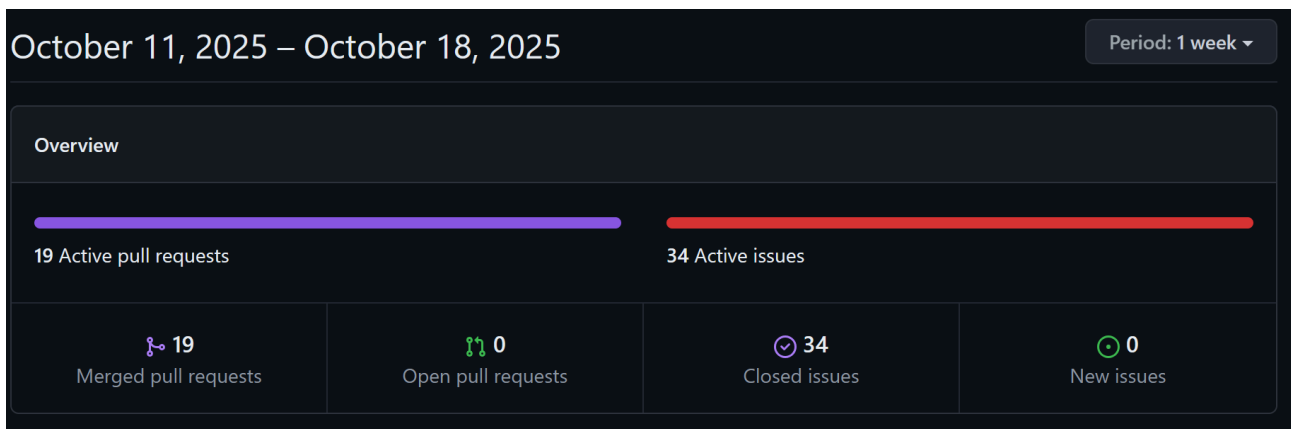
Build the Virtual Scroll Access System (VSAS): a Java (Java-17) application that lets users register/login, browse and manage “digital scrolls” (binary files). Guests browse read-only, authenticated users to upload and manage “scrolls”, and administrators to manage users and review usage. The project follows Scrum with time-boxed sprints and CI support.

### Objectives set for the project

- User Management: registration, secure login; profile updates; guest access (read-only); one admin with powers to view users, add/delete users, view stats (upload/download counts), grant roles; display user type; store passwords as a one-way hash.
- Scroll Management: add/upload (unique id and name); edit/update; remove by owner.
- Scroll Seeker: view all scrolls; download directly from the list; search filters (uploader id, scroll id, name, date); preview.
- Scrum and CI: plan and demo each sprint, maintain a project board, demo + weekly report, provide a tagged code snapshot for each demo; CI with automated tests and >75% coverage.

In Sprint-1 we met the user-management and basic scroll-management objectives via the items listed below; admin features and search/preview are scheduled for later sprints.

### Sprint-1 evidence of features implemented and objectives met



Registration (CLI): [#39](#), [#13](#)

Login/Logout and session (CLI): [#40](#), [#34](#), [#18](#), [#17](#), [#11](#)

Whoami / identity display: [#23](#), [#20](#)

Catalogue listing: [#32](#), [#20](#)

Upload (unique scroll id & name): [#34](#), [#32](#), [#30](#), [#28](#), [#21](#)

Unique user idKey enforcement: [#30](#), [#28](#), [#27](#), [#26](#), [#25](#), [#16](#)

Password hashing (one-way, salted): [#41](#), [#40](#), [#39](#), [#38](#), [#37](#), [#17](#)

Guest guard (read-only without login): [#24](#), [#23](#), [#21](#), [#20](#), [#19](#)

## 1.2 Sprint Goals

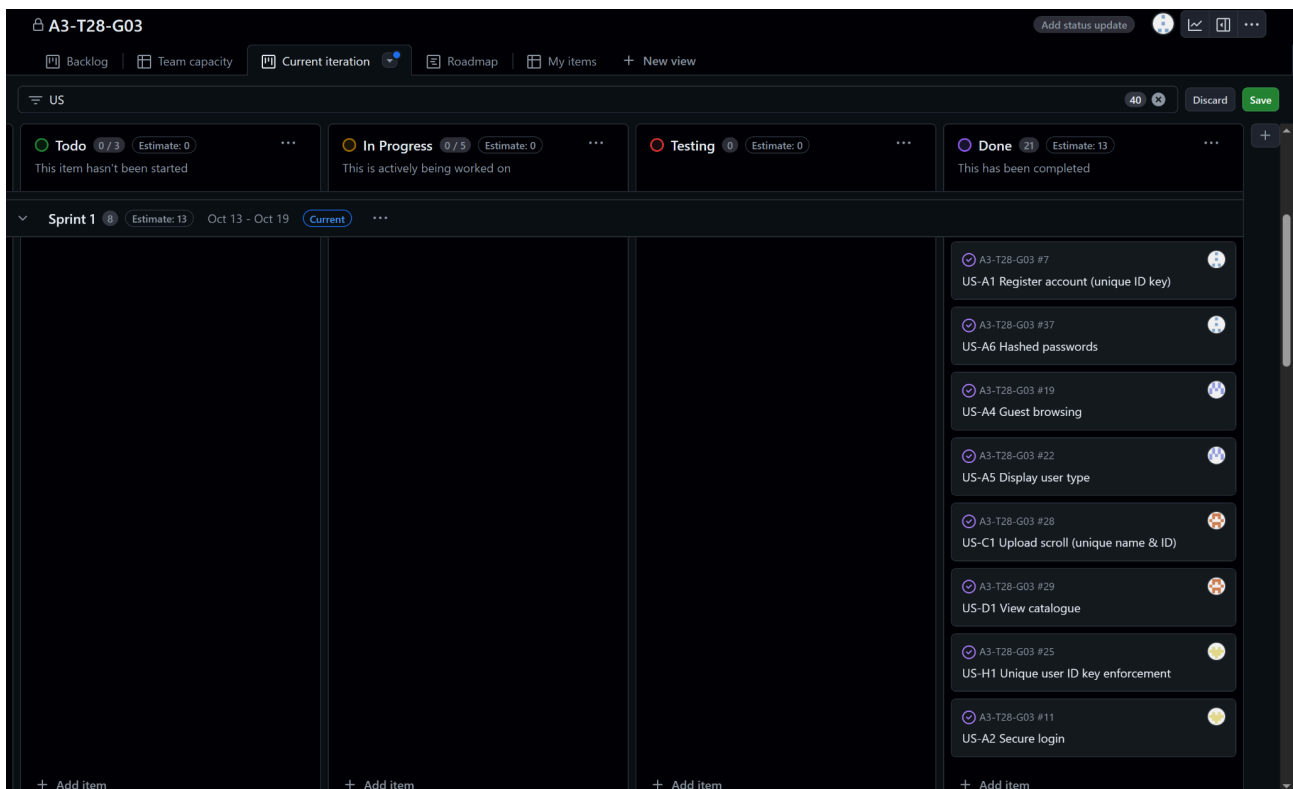
Sprint-1 goal (MVP increment). Deliver a terminal-based MVP enabling:

- Anonymous browsing (list) and user identity display (whoami);
- Account creation and secure login (with hashed passwords);
- Unique-id upload of scrolls (metadata stored; file copied), visible in catalogue;
- Enforce unique user idKey and guest-only restrictions.

Alignment to project objectives. This increment provides the first usable path for staff/members to access the library (browse and upload), while meeting security constraints (hashed passwords) and starting data integrity rules (unique ids).

**Total Sprint-1 issues:** 34/34 closed, including **8 user stories CLOSED** and **26 task issues CLOSED**.

Evidence from the project board (Sprint-1).



**User stories in Sprint-1:**

| US code | US-A1              | US-A2               | US-A4               | US-A5               | US-A6               | US-C1               | US-D1               | US-H1               |
|---------|--------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
| issue # | <a href="#">#7</a> | <a href="#">#11</a> | <a href="#">#19</a> | <a href="#">#22</a> | <a href="#">#37</a> | <a href="#">#28</a> | <a href="#">#29</a> | <a href="#">#25</a> |

## 1.3 User Stories

User stories considered in Sprint-1 (terminal UI wording).

- US-A1 Register account (CLI)  
deliverable in this sprint (tasks closed).
- US-A2 Secure login (CLI)  
deliverable in this sprint (tasks closed).
- US-A4 Guest browsing (CLI)  
deliverable in this sprint (guard + list).
- US-A5 Display user type (CLI)  
deliverable in this sprint (tasks closed).
- US-A6 Hashed passwords  
implemented via hasher + integration tests; closed.
- US-H1 Unique user ID key enforcement  
repository check & tests closed.
- US-D1 View catalogue  
model/store/command/tests closed.
- US-C1 Upload scroll  
command/file ops/tests closed.

### Changes during the sprint.

Interface choice clarified: implemented terminal-based UI (CLI) to keep the MVP focused and testable in CI. No extra GUI scope was added.

Deferrals (per Product Backlog order): admin functions (B-series), search/filters and preview (US-D3/US-D4), and delete/edit (US-C2/US-C3) are scheduled for Sprints 2-3 to preserve a vertical slice in Sprint 1.

### Changes due to requirements (post-demo, scheduled for Sprint-2).

Tutor requested interactive masked password input (no plaintext in CLI args) for registration/login. Action: update AC for US-A1 and US-A2 to require masked prompts (and confirmation on register) and implement in Sprint-2.

## 2. Scrum Artefacts and Methodologies

### 2.1 Sprint Planning

Sprint Planning was held at the start of Week 10 with all members present.

**Sprint Date:** Week 10

**Sprint Duration:** 1 week

**Sprint Goal:**

Deliver a functional prototype of the *Virtual Scroll Access System (VSAS)* that supports secure user registration, login, and basic scroll browsing and uploading operations.

**Boning Zhao** (*Product Owner*), was responsible for defining the project scope and creating the initial set of user stories based on the client's requirements. Boning also established the initial project scaffold, including Gradle configuration and the basic testing framework. During this sprint, Boning implemented two key stories: *Secure Login* and *Unique User ID Key Enforcement*, ensuring that authentication and user-identity management formed a reliable foundation for later development.

**Xuejian Fang** (*Scrum Master*). In addition to implementing the *Register Account* and *Hashed Password* stories, Xuejian configured and deployed the Jenkins continuous-integration pipeline. This ensured that each code submission triggered an automatic build and test process, maintaining code quality and consistency. Xuejian also organised the Scrum meetings, verified that team activities followed the Scrum methodology, and facilitated communication between members.

**Xuanhao Yu** and **Zhiwei Cheng** (*development team members*). Xuanhao was responsible for implementing *Guest Browsing* and *Display User Type*, enabling non-logged-in users to view available scrolls and correctly displaying user roles within the interface. Zhiwei focused on *Upload Scroll* and *View Catalogue*, developing the core logic for file storage, retrieval, and tabular display of scroll metadata. Both developers contributed to refining the project's architecture, optimising test coverage, and improving overall code maintainability.

## 2.2 Daily Stand-ups

All stand-up meetings for this sprint were conducted **in person** to improve communication efficiency and facilitate direct collaboration. Two stand-ups were held during the week — on **Tuesday and Wednesday** — and both were led by the *Scrum Master* (**Xuejian Fang**). We sent some important messages in Wechat so that we can keep them and also used as evidence. The wechat message instances can be found in 5.2.

The **Tuesday stand-up** marked the official start of the project. Under the Scrum Master's facilitation, the **Product Owner (Boning Zhao)** drafted the initial set of user stories and assigned responsibilities to each team member. The team collaboratively designed the overall project and testing scaffold, setting up Gradle and JUnit. During the discussion, there was a brief disagreement regarding whether to include a graphical user interface or to demonstrate the system purely through a terminal interface. After collective deliberation, the team agreed to use a **terminal-based demo** for this sprint and reserve GUI design for future iterations.

The **Wednesday stand-up** focused on reviewing each member's progress and merging completed story implementations. Team members presented their individual features, which were integrated into the main branch. The Scrum Master executed the **CI/CD pipeline** using Jenkins and refined the build configuration to ensure successful automated testing and deployment. Several issues arose during this process — including code-merge conflicts and minor functional inconsistencies — but these were resolved collaboratively through quick debugging and pair-programming sessions.

## 2.3 Retrospective

The sprint retrospective meeting was held at the end of Week 10, with all team members present. The primary goal of the meeting was to evaluate the overall outcomes of this sprint, identify issues

encountered during development and collaboration, and propose improvements for the next iteration.

In terms of achievements, the team maintained strong in-person collaboration and efficient communication throughout the sprint. The core functionalities for user registration and authentication were successfully implemented. The Jenkins CI/CD pipeline was configured during this period and was able to automatically execute builds and tests after each code submission. All planned user stories were completed within the expected timeframe and successfully demonstrated to the client.

Despite these successes, several challenges were identified. Multiple merge conflicts occurred during the integration stage due to overlapping file edits. Some test cases initially failed because of inconsistent mock data formats. Communication regarding branch naming and merging responsibilities also required further clarification to ensure smoother coordination.

To address these issues, the team decided to establish clearer branch ownership and review procedures for the next sprint, performing unified validation before merging. Additional automated tests for boundary conditions and data validation will be introduced, with the goal of increasing overall test coverage to above 80%. The team also agreed to continue conducting in-person stand-ups but to fix regular meeting times on Tuesdays and Fridays to maintain stable coordination.

The tutor provided positive feedback during the sprint review, acknowledging that the current system meets the basic functional requirements. The tutor encouraged the team to extend the scroll management module—particularly the search and filtering functions—and to begin planning for the integration of a graphical user interface (GUI) in the next sprint.

## 2.4 Use of Scrum Artefacts

Project backlog and Sprint backlog:

<https://github.sydney.edu.au/orgs/SOFT2412-COMP9412-2025s2/projects/309/views/1>

The Jenkins Console Output:

[https://github.sydney.edu.au/SOFT2412-COMP9412-2025s2/A3-T28-G03/blob/Appendix\\_A/Jenkins\\_Console\\_output.png](https://github.sydney.edu.au/SOFT2412-COMP9412-2025s2/A3-T28-G03/blob/Appendix_A/Jenkins_Console_output.png)

During this sprint, the team managed and maintained all core Scrum artefacts — the **Product Backlog**, **Sprint Backlog**, and **User Stories** — through the **GitHub Projects board**. The Product Owner created and organised a comprehensive list of user stories based on client requirements, which together formed the Product Backlog. These stories were documented in the *Stories* column of the board.

During the sprint planning meeting, the team selected the highest-priority items to constitute the Sprint Backlog for the current iteration. Tasks were distributed among members and tracked across the *To Do*, *In Progress*, *Testing*, and *Done* columns to visualise the workflow. Each user story contained a unique identifier (e.g., US-A1, US-C1) with defined priorities, estimates, and assignees, ensuring clarity and traceability of responsibilities.

The Scrum Master updated task statuses after each stand-up meeting to maintain visibility and alignment across the team. This shared dashboard enabled all members to monitor progress in real time, supporting continuous integration and collaborative development throughout the sprint.

The complete Product Backlog, Sprint Backlog, and User Stories are accessible through the team's GitHub Projects board at the following link:

<https://github.sydney.edu.au/orgs/SOFT2412-COMP9412-2025s2/projects/309/views/1>

Additionally, the team's CI/CD automation and build verification were handled through Jenkins. The Jenkins console output screenshot is included in **Appendix A**, demonstrating successful automatic build, testing, and validation after each code submission. This integration ensured consistent software quality and reliable delivery throughout the sprint.

## 3. Agile Development Tools and Practices

### 3.1 Version Control

#### 3.1.1 Tools and Branching Strategy

We use Git and GitHub for source control, Issues, Projects (backlog), and Pull Requests. Jenkins provides automated CI based on the repository Jenkinsfile, and Gradle handles build/test with google-java-format for consistent style. Our branching model is: a protected main branch (no direct commits, merges only via PRs), feature branches per User Story named feature/<story-name> (e.g., feature/us-a6-hashed-passwords, feature/us-a1-register-idkey), and small, incremental commits per Task using conventional prefixes (feat/fix/test/refactor/docs/chore) that reference issue IDs. Non-story fixes and chores use dedicated prefixes (e.g., fix/, chore/) and are also merged via PRs. A PR is raised only after all Tasks for the Story are complete and Acceptance Criteria are met; related Issues are linked in the PR's Development section; CI must be green before merge (we prefer squash merge).

#### 3.1.2 Merge Conflicts and Branch Integration

To prevent conflicts, each feature branch regularly syncs with main (git pull --rebase origin/main, or merge if preferred) and runs local tests and formatting before pushing. If a conflict arises, we resolve it locally with two guardrails: preserve public contracts (e.g., UserRepository and PasswordHasher signatures) and maintain separation of concerns (repository handles TSV I/O and headers; CLI handles parsing/validation/orchestration). After resolution we run the full test suite and formatting, explain decisions in the commit or PR description (for example, keeping a 3-arg constructor for backward compatibility while adding a 4-arg overload for dependency injection), wait for Jenkins to pass, review, and squash-merge. No conflicts occurred in this iteration; if they occur in later sprints, we will document the actual steps and outcomes in the report.

#### 3.1.3 How Added Features Fit User Requirements

We begin from User Stories (role, goal, value), derive Tasks and Acceptance Criteria, and implement features directly against those criteria with tests and CLI demos.

Example: US-A1 "Register account (unique ID key)". Story: "As a Visitor, I want to register via CLI so that I can upload and manage scrolls." Acceptance Criteria: running `register --username <u> --password <p> --email <e> --phone <ph> --id-key <k>` exits 0 with a success message and <k> is unique; if <k> already exists, exit is non-zero with a clear error. Our implementation aligns as follows: RegisterCommand parses the five required flags, checks uniqueness via UserRepository.existsIdKey (duplicate fails early without appending to TSV), and never stores plaintext passwords. PasswordHasher generates a per-user SecureRandom salt and computes SHA-256(salt || UTF-8(password)) to a hex digest; constant-time comparison is used in authentication. FileUserRepository ensures data/users.tsv has a header, performs atomic appends, and provides find/save/exists operations. Automated tests and CLI runs cover success, duplicate idKey, missing fields, and "no plaintext in TSV," and Jenkins enforces these checks on every PR.

3.14 Based on user acceptance in Sprint 1, beyond the salted-hash protection already implemented, users requested stronger security: when using the registration command, the password input should be hidden (no echo) while typing.





Filters ▾

is:issue is:closed

Labels 9

Milestones 0

New issue

✕ Clear current search query, filters, and sorts

9 Open ✓ 25 Closed

Author ▾

Label ▾

Projects ▾

Milestones ▾

Assignee ▾

Sort ▾

|                          |                                                                                                                                                                                              |   |  |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|--|
| <input type="checkbox"/> | Unit tests: stores non-plaintext; correct vs wrong password; distinct salts lead to distinct hashes for the same password.<br>#41 by bzha8676 was closed 3 hours ago                         | 1 |  |
| <input type="checkbox"/> | Integrate verification in <code>login</code> : load salt → recompute hash → constant-time compare.<br>#40 by bzha8676 was closed 3 hours ago                                                 | 1 |  |
| <input type="checkbox"/> | Integrate hashing in <code>register</code> : generate salt → hash → persist <code>passwordHash</code> + <code>salt</code> (never store plaintext).<br>#39 by bzha8676 was closed 3 hours ago | 1 |  |
| <input type="checkbox"/> | Implement <code>PasswordHasher</code> (salted SHA-256 using JDK <code>MessageDigest</code> ; per-user random 16–32 bytes salt; hex or Base64).<br>#38 by bzha8676 was closed 3 hours ago     | 1 |  |
| <input type="checkbox"/> | Tests: success / duplicate id / not logged in / file missing<br>#36 by bzha8676 was closed 3 hours ago                                                                                       |   |  |
| <input type="checkbox"/> | File ops: copy to <code>data/files/&lt;sid&gt;.bin</code> ; write metadata; reject duplicate <code>&lt;sid&gt;</code><br>#35 by bzha8676 was closed 3 hours ago                              |   |  |
| <input type="checkbox"/> | Command: <code>vsas upload --id &lt;sid&gt; --name "&lt;n&gt;" --file &lt;path&gt;</code> (login required)<br>#34 by bzha8676 was closed 3 hours ago                                         |   |  |
| <input type="checkbox"/> | Tests: empty library message / single & multiple rows<br>#33 by bzha8676 was closed 3 hours ago                                                                                              |   |  |
| <input type="checkbox"/> | Command: <code>vsas list</code> → print table: id   name   uploader   uploadDate<br>#32 by bzha8676 was closed 3 hours ago                                                                   |   |  |
| <input type="checkbox"/> | Storage: <code>FileScrollRepository</code> with TSV <code>data/scrolls.tsv</code><br>#31 by bzha8676 was closed 3 hours ago                                                                  |   |  |
| <input type="checkbox"/> | Model: create <code>Scroll</code> (id, name, uploaderIdKey, uploadDate, filePath, downloadCount)<br>#30 by bzha8676 was closed 3 hours ago                                                   |   |  |
| <input type="checkbox"/> | Tests: duplicate idKey rejected (also concurrent add simulated by repeated calls)<br>#27 by bzha8676 was closed 3 hours ago                                                                  |   |  |
| <input type="checkbox"/> | Repo check: fast lookup for <code>idKey</code> in <code>FileUserRepository</code><br>#26 by bzha8676 was closed 3 hours ago                                                                  |   |  |
| <input type="checkbox"/> | Command: <code>vsas whoami</code> → prints <code>guest</code> or <code>username</code> (role=user admin)<br>#23 by bzha8676 was closed 3 hours ago                                           |   |  |
| <input type="checkbox"/> | Tests: guest blocked on <code>upload</code> (and future <code>download</code> )<br>#21 by bzha8676 was closed 3 hours ago                                                                    |   |  |
| <input type="checkbox"/> | Guard: shared check so guest can only run <code>list</code> / <code>whoami</code><br>#20 by bzha8676 was closed 3 hours ago                                                                  |   |  |
| <input type="checkbox"/> | Tests: valid login / invalid login / logout clears session<br>#18 by bzha8676 was closed 3 hours ago                                                                                         |   |  |
| <input type="checkbox"/> | Command: <code>vsas login --username &lt;u&gt; --password &lt;p&gt;</code> (verify hash) / <code>vsas logout</code><br>#17 by bzha8676 was closed 5 hours ago                                |   |  |
| <input type="checkbox"/> | Session: <code>SessionService</code> with file <code>\${user.home}/.vsas/session.properties</code> (username, idKey, role, issuedAt)<br>#16 by bzha8676 was closed 3 hours ago               |   |  |
| <input type="checkbox"/> | Tests: success / duplicate idKey / missing fields / password not stored in plain text<br>#15 by bzha8676 was closed 12 hours ago                                                             | 1 |  |
| <input type="checkbox"/> | Validation: enforce unique <code>idKey</code> (fail with non-zero exit code)<br>#14 by bzha8676 was closed 12 hours ago                                                                      | 1 |  |
| <input type="checkbox"/> | Command: <code>vsas register --username &lt;u&gt; --password &lt;p&gt; --email &lt;e&gt; --phone &lt;ph&gt; --id-key &lt;k&gt;</code><br>#13 by bzha8676 was closed 12 hours ago             | 1 |  |
| <input type="checkbox"/> | Hasher: <code>PasswordHasher</code> (salted SHA-256, hex encode)<br>#10 by bzha8676 was closed 12 hours ago                                                                                  | 1 |  |
| <input type="checkbox"/> | Storage: create <code>FileUserRepository</code> using TSV file <code>data/users.tsv</code> (header + tab-separated)<br>#9 by bzha8676 was closed 12 hours ago                                | 1 |  |
| <input type="checkbox"/> | Model: create <code>User</code> (username, email, phone, idKey, role, passwordHash, salt, createdAt)<br>#8 by bzha8676 was closed 12 hours ago                                               | 1 |  |

ProTip! Find all open issues with in progress development work with [linked:pr](#).

Code

Issues

Pull requests

Discussions

Projects

Wiki

Security

Insights

Settings

A3-T28-G03

Public

Edit Pins

Watch

Fork

Starred

main

22 Branches

0 Tags

Go to file

Add file

Code

bzha8676

Merge pull request #52 from SOFT2412-COMP9412-2025a2/fix/c...

6778b1d · 6 hours ago

72 Commits

app

fix(cli): align usage text with commands

6 hours ago

gradle

chore: initialize project via Gradle init and add CI

2 days ago

.gitattributes

chore: initialize project via Gradle init and add CI

2 days ago

.gitignore

add "data" folder in gitignore

7 hours ago

Jenkinsfile

ci: re-enable googleJavaFormat check with non-blocking ...

yesterday

LICENSE

Initial commit

last week

README.md

fix(docs): update test report paths in README

6 hours ago

gradle.properties

chore: add gradle.properties to configure console output

12 hours ago

gradlew

fix(cli): make gradlew executable for CI and local usage

yesterday

gradlew.bat

chore: initialize project via Gradle init and add CI

2 days ago

settings.gradle

chore: initialize project via Gradle init and add CI

2 days ago

README

MIT license

CC28-Team03-A2A3 - VSAS

Run

.gradlew run

Test & Coverage

.gradlew test jacocoTestReport

Reports: `app/build/reports/tests/test`, `app/build/reports/jacoco/test/html`

Code Style (Google Java Format)

Format locally: `./gradlew googleJavaFormat`

Verify in CI (already wired to `check`): `./gradlew verifyGoogleJavaFormat`

About

No description, website, or topics provided.

Readme

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Activity

Custom properties

2 stars

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Releases

No releases published

Create a new release

Contributors

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Xuanhao Yu

Languages

Java

100.0%

Clear current search query, filters, and sorts

| <input type="checkbox"/> | 0 Open <input checked="" type="checkbox"/> 18 Closed                                                                                             | Author | Label | Projects | Milestones | Reviews | Assignee | Sort |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|--------|-------|----------|------------|---------|----------|------|
| <input type="checkbox"/> | <b>Fix/cli usage help text</b> ✓<br>#52 by bzha8676 was merged 6 hours ago                                                                       |        |       |          |            |         |          |      |
| <input type="checkbox"/> | <b>add "data" folder in gitignore</b> ✓<br>#51 by xuyu8020 was merged 7 hours ago                                                                |        |       |          |            |         |          |      |
| <input type="checkbox"/> | <b>feat(cli): implement session persistence in Login and Logout commands</b> ✓<br>#50 by bzha8676 was merged 11 hours ago                        |        |       |          |            |         |          |      |
| <input type="checkbox"/> | <b>Feature/interactive shell</b> ✓<br>#49 by bzha8676 was merged 12 hours ago                                                                    |        |       |          |            |         |          |      |
| <input type="checkbox"/> | <b>US-A1: CLI registration with TSV storage and unique idKey (A3-T28-G03 #7)</b> ✓<br>#48 by xfan0282 was merged 14 hours ago                    |        |       |          |            | 7       |          |      |
| <input type="checkbox"/> | <b>feat(scroll): add list command with TSV-backed repositor</b> ✗<br>#47 by zche0826 was merged yesterday                                        |        |       |          |            |         |          |      |
| <input type="checkbox"/> | <b>chore(format): re-run googleJavaFormat for RegisterCommand.java and A...</b> ✓<br>#46 by xfan0282 was merged yesterday                        |        |       |          |            |         |          |      |
| <input type="checkbox"/> | <b>US-A6: Hashed passwords — implement salted SHA-256, register hashing, login verification, tests</b> ✓<br>#45 by xfan0282 was merged yesterday |        |       |          |            | 5       |          |      |
| <input type="checkbox"/> | <b>Feat/cli implement upload command and correct commit history</b> ✓<br>#44 by zche0826 was merged yesterday                                    |        |       |          |            |         |          | 1    |
| <input type="checkbox"/> | <b>Chore/format google java</b> <b>invalid</b><br>#43 by bzha8676 was closed yesterday                                                           |        |       |          |            |         |          |      |
| <input type="checkbox"/> | <b>feat(cli): update CommandDispatcher to return success on usage help a...</b><br>#42 by zche0826 was merged yesterday                          |        |       |          |            |         |          |      |
| <input type="checkbox"/> | <b>chore(cli): add runnable CLI skeleton and package structure</b><br>#12 by bzha8676 was merged yesterday                                       |        |       |          |            |         |          |      |
| <input type="checkbox"/> | <b>Fix/ci gradlew and format check</b><br>#6 by xfan0282 was merged yesterday                                                                    |        |       |          |            |         |          |      |
| <input type="checkbox"/> | <b>Chore/configure project tooling</b><br>#5 by bzha8676 was merged yesterday                                                                    |        |       |          |            |         |          |      |
| <input type="checkbox"/> | <b>Fix/ci jenkinsfile fix</b><br>#4 by xfan0282 was merged 2 days ago                                                                            |        |       |          |            |         |          |      |
| <input type="checkbox"/> | <b>chore: initialize project via Gradle init and add CI</b><br>#3 by bzha8676 was merged 2 days ago                                              |        |       |          |            |         |          |      |
| <input type="checkbox"/> | <b>docs: add README with run/test instructions and project .gitignore</b><br>#2 by bzha8676 was merged 2 days ago                                |        |       |          |            |         |          |      |
| <input type="checkbox"/> | <b>ci(jenkins): add Jenkinsfile for Gradle multi-branch pipeline</b> ✗<br>#1 by xfan0282 was merged 2 days ago                                   |        |       |          |            |         |          |      |

ProTip! Click a checkbox on the left to edit multiple issues at once.

Branches

New branch

Overview   Yours   Active   Stale   All

Q Search branches...

| Branch                                                      | Updated      | Check status | Behind | Ahead | Pull request        |     |
|-------------------------------------------------------------|--------------|--------------|--------|-------|---------------------|-----|
| <code>fix/cli-usage-help-text</code>                        | 7 hours ago  | ✓ 1/1        | 1      | 0     | <a href="#">#52</a> | ... |
| <code>chore/gitignore-data</code>                           | 7 hours ago  | ✓ 1/1        | 5      | 0     | <a href="#">#51</a> | ... |
| <code>feature/cli-session</code>                            | 11 hours ago | ✓ 1/1        | 7      | 0     | <a href="#">#50</a> | ... |
| <code>feature/interactive-shell</code>                      | 12 hours ago | ✓ 1/1        | 16     | 0     | <a href="#">#49</a> | ... |
| <code>feature/us-a1-Register-account</code>                 | 14 hours ago | ✓ 1/1        | 14     | 0     | <a href="#">#48</a> | ... |
| <code>feat/US-A4-guest-browsing</code>                      | 19 hours ago | ✓ 1/1        | 18     | 1     |                     | ... |
| <code>feat/US-A5-display-user-type</code>                   | 20 hours ago | ✓ 1/1        | 18     | 0     |                     | ... |
| <code>feat/scroll-list-command</code>                       | yesterday    | ✗ 0/1        | 49     | 0     | <a href="#">#47</a> | ... |
| <code>fix/formatting-registercommand-authhashingtest</code> | yesterday    | ✓ 1/1        | 21     | 0     | <a href="#">#46</a> | ... |
| <code>feature/us-a6-hashed-passwords</code>                 | yesterday    | ✓ 1/1        | 48     | 0     | <a href="#">#45</a> | ... |
| <code>feat/cli-implement-upload-command</code>              | yesterday    | ✓ 1/1        | 28     | 0     | <a href="#">#44</a> | ... |
| <code>feat/scroll/cli-upload-unique-id</code>               | yesterday    | ✗ 0/1        | 51     | 0     |                     | ... |
| <code>fix/ci-jenkinsfile-fix</code>                         | yesterday    | ✓ 1/1        | 63     | 0     |                     | ... |
| <code>fix/ci-gradlew-and-format-check</code>                | yesterday    | ✓ 1/1        | 55     | 0     |                     | ... |
| <code>feature/sprint1-cli-skeleton</code>                   | yesterday    | ✓ 1/1        | 53     | 0     |                     | ... |
| <code>docs/readme-and-gitignore</code>                      | yesterday    | ✗ 0/1        | 68     | 0     |                     | ... |
| <code>chore/configure-project-tooling</code>                | yesterday    | ✓ 1/1        | 58     | 0     |                     | ... |
| <code>chore/bootstrap-with-gradle-init</code>               | yesterday    | ✗ 0/1        | 66     | 0     |                     | ... |
| <code>feature/add-jenkinsfile</code>                        | 2 days ago   | ✗ 0/1        | 70     | 0     | <a href="#">#1</a>  | ... |
| <code>feature/US-H1-unique-userid-enforcement</code>        | 2 days ago   |              | 71     | 0     |                     | ... |

SOFT2412-COMP9412-2025s2 / Projects / A3-T28-G03

Q Type / to search

A3-T28-G03

BacklogTeam capacityCurrent iterationRoadmap

Filter by keyword or by field

Stories27Estimate: 13

A3-T28-G03 #7

US-A1 Register account (unique ID key)

P01M

A3-T28-G03 #19

US-A4 Guest browsing

P02M

A3-T28-G03 #22

US-A5 Display user type

P01S

A3-T28-G03 #28

US-C1 Upload scroll (unique name & ID)

P02M

A3-T28-G03 #29

US-D1 View catalogue

P02L

A3-T28-G03 #11

US-A2 Secure login

P02M

A3-T28-G03 #37

US-A6 Hashed passwords

P12M

A3-T28-G03 #25

+ Add item

Todo6/38Estimate: 0

This item hasn't been started

A3-T28-G03 #7

US-A1 Register account (unique ID key) #7

Openbzza8676 opened yesterday

bzza8676 12 hours ago (edited)

Edit

Story

As a Visitor, I want to register via CLI, So that I can upload and manage scrolls.

Acceptance Criteria

- When I run `visas register --username <u> --password <p> --email <e> --phone <ph> --id-key <id>` exit code is 0 and a success message prints; the `<id>` is unique.
- If `<id>` already exists, the command fails with non-zero exit code and a clear error.

Tasks

- ☒ Model: create `User` (username, email, phone, idKey, role, passwordHash, salt, createdAt) #8
- ☒ Storage: create `FileUserRepository` using TSV file `@data/users.tsv` (header + tab-separated) #9
- ☒ Hasher: `PasswordHasher` (salted SHA-256, hex encode) #10
- ☒ Command: `visas register --username <u> --password <p> --email <e> --phone <ph> --id-key <id>` #13
- ☒ Validation: enforce unique `idKey` (fail with non-zero exit code) #14
- ☒ Tests: success / duplicate idKey / missing fields / password not stored in plain text #15

Commit msg: feat(auth): add CLI registration with salted hashing and TSV storage

WritePreview

H B I

Leave a comment

Paste, drop, or click to add files

Close issueComment

Assignees

Labels

Milestone

Status

Linked pull requests

Repository

Priority

Size

Estimate

Sprint

Start date

End date

Open in new tab

Copy link

Copy link in project

Archive

Delete from project

US-A1: CLI registration with TSV storage and unique idKey (A3-T28-G03 #7) #48

Edit

Merged

xfan0282 merged 6 commits into main from feature/us-a1-register-account 14 hours ago

Conversation 0

Commits 6

Checks 0

Files changed 9

+564 -75

xfan0282 commented 14 hours ago

Member

Summary

Implement CLI user registration with salted hashing and TSV-backed persistence. Covers A3-T28-G03 #7 via tasks #8-#15.

- model/User: add fields (username, email, phone, idKey, role, passwordHash, salt, createdAt). (Model: create User (username, email, phone, idKey, role, passwordHash, salt, createdAt) #8)
- repo/FileUserRepository: TSV data/users.tsv with header; read/write helpers; existsIdKey. (Storage: create FileUserRepository using TSV file data/users.tsv (header + tab-separated) #9)
- security/PasswordHasher: SHA-256(salt|UTF-8(pwd)), hex; per-user SecureRandom salt; const-time compare. (reused, Hasher: PasswordHasher (salted SHA-256, hex encode) #10)
- cli/RegisterCommand: parse required flags --username --password --email --phone --id-key; hash+salt and persist via repo; success message on exit 0. (Command: vsas register --username <u> --password <p> --email <e> --phone <ph> --id-key <k> #13)
- cli/RegisterCommand: enforce unique idKey; duplicate → non-zero exit with clear error. (Validation: enforce unique idKey (fail with non-zero exit code) #14)
- tests: success path, duplicate idKey, missing fields, and no plaintext password in TSV. (Tests: success / duplicate idKey / missing fields / password not stored in plain text #15)

Linked Issues

Story: A3-T28-G03 #7 · Tasks: #8 #9 #10 #13 #14 #15

xfan0282 added 6 commits yesterday

feat(model): add User.createdAt and backward-compatible constructor (...)

feat(repo): implement TSV-backed FileUserRepository with minimal test...

feat(security): add salted SHA-256 password hasher with tests (A3-T28...

feat(cli): implement vsas register with salted hashing via repository...

feat(cli): enforce unique idKey in register (A3-T28-G03#14)

test(cli): cover register success/duplicate/missing and no-plaintext ...

xfan0282 self-assigned this 14 hours ago

This was linked to issues 14 hours ago

US-A1 Register account (unique ID key) #7

Model: create User (username, email, phone, idKey, role, passwordHash, salt, createdAt) #8

Storage: create FileUserRepository using TSV file data/users.tsv (header + tab-separated) #9

Hasher: PasswordHasher (salted SHA-256, hex encode) #10

Command: vsas register --username <u> --password <p> --email <e> --phone <ph> --id-key <k> #13

Validation: enforce unique idKey (fail with non-zero exit code) #14

Tests: success / duplicate idKey / missing fields / password not stored in plain text #15

xfan0282 merged commit c85717a into main 14 hours ago

Revert

1 participant

Lock conversation

Notifications

Customize

Unsubscribe

You're receiving notifications because you modified the open/close state.

US-A1 Register account (unique ID key)

Command: vsas register --username <u> ...

Hasher: PasswordHasher (salted SHA-256, h...

Storage: create FileUserRepository using T...

Model: create User (username, email, phon...

See more

Pull request successfully merged and closed

Delete branch

11/48/commits

## 3.2 CI/CD Tools and Practices

### 3.2.1 CI Pipeline (Jenkins)

We use Jenkins Multibranch Pipeline as the primary CI. The pipeline is defined in the repository Jenkinsfile and executes: Checkout → Sanity → Format (google-java-format verify) → Build (Gradle) → Test (JUnit + JaCoCo) → Archive (versioned JAR, fingerprinted) → Deploy (placeholder on main only). Recent builds are green with tests passing and artifacts archived; typical turnaround is within seconds to a minute, enabling fast feedback.



### 3.2.2 Status Visibility in GitHub

Each commit and pull request shows the Jenkins check on the right side in GitHub. When CI passes, a green check appears; if it fails, the PR is blocked from merging until resolved. This gives clear, enforceable quality gates at the PR level.

### 3.2.3 Triggers and Branch Discovery

Jenkins integrates with GitHub using a configured GitHub server and credentials. The Multibranch job discovers feature/\* branches and pull requests automatically and builds both the PR head and the “PR merged into target” variant. This supports our protected-branch workflow and ensures PRs are validated in the context of the target branch.

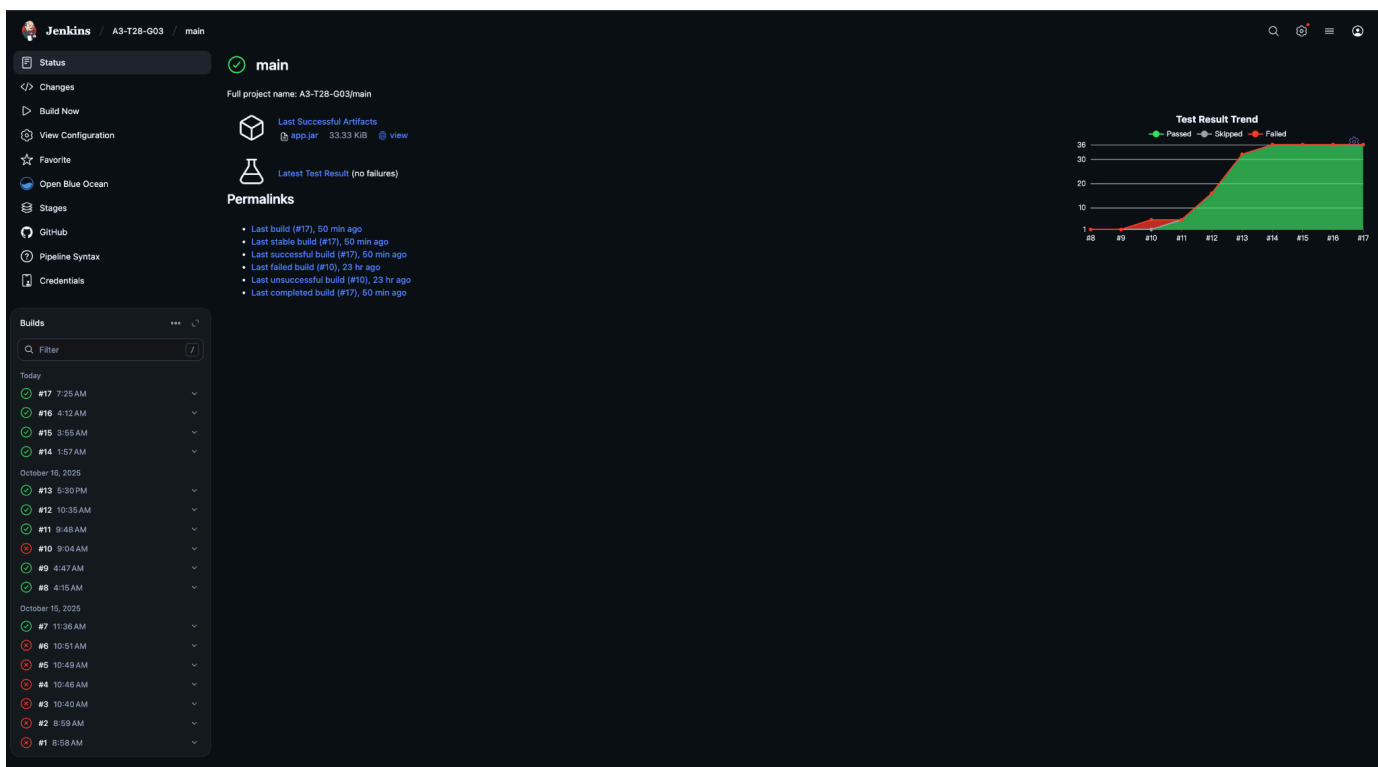
### 3.2.4 CD Status and Next Steps

Current scope is Continuous Integration only; Continuous Delivery is not yet implemented. The Deploy stage in the Jenkinsfile is a placeholder and does not update any environment. In upcoming iterations we plan to add CD to a test/pre-prod environment (publish artifact, smoke tests, rollback strategy) and later consider Continuous Deployment once sufficient confidence is achieved.

The screenshot shows the Jenkins web interface for job A3-T28-G03. The left sidebar contains navigation links for Status, Configure, Scan Repository Now, Scan Repository Log, Multibranch Pipeline Events, Delete Multibranch Pipeline, Build History, Project Relationship, Check File Fingerprint, Favorite, Open Blue Ocean, GitHub, Rename, Pipeline Syntax, and Credentials. The main panel displays a table of build jobs with columns for status (S), warning (W), name, last success, last failure, last duration, and a flag (F). The table lists 20 branches, including chore/bootstrap-with-gradle-init, chore/configure-project-tooling, chore/gitignore-data, docs/readme-and-gitignore, feat/ci-implement-upload-command, feat/scroll/ci-upload-unique-id, feat/scroll-list-command, feat/US-A4-guest-browsing, feat/US-A5-display-user-type, feature/add-jenkinsfile, feature/ci-session, feature/interactive-shell, feature/hprint1-cli-skeleton, feature/ua-v1-register-account, feature/ua-a6-hashed-passwords, fix/ci-gradlew-and-format-check, fix/ci-jenkinsfile-fix, fix/ci-usage-help-text, fix/formatting-registercommand-authhashingtest, and main. The status of each branch is indicated by a green checkmark (passing) or a red X (failing). The last success and last failure columns show the time and build number of the most recent successful or failed build. The last duration column shows the time taken for the most recent build. The flag column contains a star icon for each branch.

| S | W | Name                                           | Last Success | Last Failure   | Last Duration | F |
|---|---|------------------------------------------------|--------------|----------------|---------------|---|
| ✓ |   | chore/bootstrap-with-gradle-init               | N/A          | 50 min #5      | 3 min 0 sec   | ★ |
| ✓ |   | chore/configure-project-tooling                | 50 min #3    | N/A            | 1 min 40 sec  | ★ |
| ✓ |   | chore/gitignore-data                           | 50 min #2    | N/A            | 32 sec        | ★ |
| ✓ |   | docs/readme-and-gitignore                      | N/A          | 50 min #3      | 2 min 58 sec  | ★ |
| ✓ |   | feat/ci-implement-upload-command               | 50 min #2    | N/A            | 46 sec        | ★ |
| ✓ |   | feat/scroll/ci-upload-unique-id                | N/A          | 50 min #4      | 3 min 14 sec  | ★ |
| ✓ |   | feat/scroll-list-command                       | N/A          | 50 min #2      | 2 min 55 sec  | ★ |
| ✓ |   | feat/US-A4-guest-browsing                      | 50 min #2    | N/A            | 2 min 30 sec  | ★ |
| ✓ |   | feat/US-A5-display-user-type                   | 50 min #2    | N/A            | 57 sec        | ★ |
| ✓ |   | feature/add-jenkinsfile                        | N/A          | 50 min #3      | 1 min 39 sec  | ★ |
| ✓ |   | feature/ci-session                             | 50 min #2    | N/A            | 1 min 11 sec  | ★ |
| ✓ |   | feature/interactive-shell                      | 50 min #2    | N/A            | 2 min 50 sec  | ★ |
| ✓ |   | feature/hprint1-cli-skeleton                   | 50 min #3    | N/A            | 2 min 25 sec  | ★ |
| ✓ |   | feature/ua-v1-register-account                 | 50 min #2    | N/A            | 1 min 38 sec  | ★ |
| ✓ |   | feature/ua-a6-hashed-passwords                 | 50 min #7    | N/A            | 2 min 5 sec   | ★ |
| ✓ |   | fix/ci-gradlew-and-format-check                | 50 min #3    | N/A            | 2 min 2 sec   | ★ |
| ✓ |   | fix/ci-jenkinsfile-fix                         | 50 min #4    | 1 day 20 hr #1 | 21 sec        | ★ |
| ✓ |   | fix/ci-usage-help-text                         | 50 min #2    | N/A            | 3 min 33 sec  | ★ |
| ✓ |   | fix/formatting-registercommand-authhashingtest | 50 min #2    | N/A            | 3 min 21 sec  | ★ |
| ✓ |   | main                                           | 50 min #17   | 23 hr #10      | 1 min 23 sec  | ★ |





## Webhooks

Add webhook

Webhooks allow external services to be notified when certain events happen. When the specified events happen, we'll send a POST request to each of the URLs you provide. Learn more in our [Webhooks Guide](#).

We will also send events from this repository to your [organization webhooks](#).

✓ <https://kscii.tech/github-webhook/> (pull\_request and push)

Edit

Delete

Branch Sources

≡

GitHub

✕

Credentials ?

xfan0282/\*\*\*\*\*

▼

+ Add

☒ Repository HTTPS URL

Repository HTTPS URL ?

https://github.sydney.edu.au/SOFT2412-COMP9412-2025s2/A3-T28-G03

Validate

☐ Repository Scan - Deprecated Visualization

Behaviours

Discover branches ?

Strategy ?

All branches

▼

Discover pull requests from origin ?

Strategy ?

Both the current pull request revision and the pull request merged with the current target branch revision

▼

Add ▼

Property strategy

Named branches get different properties

▼

Exceptions

Add exception

Defaults

Add property ▼

### 3.3 Code Quality

### 3.3.1 Tooling and Linting

We standardise formatting with google-java-format (Gradle plugin). Developers must run `./gradlew googleJavaFormat` before committing. Jenkins also runs `verifyGoogleJavaFormat` on every build to catch drift; the stage is visible in the pipeline log. Beyond formatting, we use Gradle + JUnit 5 for tests and JaCoCo to generate coverage reports on every CI run (`jacocoTestReport`). GitHub shows a green check on commits and PRs when CI passes.

### 3.3.2 Tests and Coverage

Unit tests cover CLI commands, repositories, the session service, and password hashing. The latest report shows 32 tests, 100% passing. Overall instruction coverage is approximately 83% (with package-level coverage such as cli  $\approx$ 81%, repo  $\approx$ 87%, service  $\approx$ 71%, security  $\approx$ 91%, model  $\approx$ 87%). These reports are archived by Jenkins for every build, providing a consistent baseline for regressions.

### 3.3.3 Code Reviews

A formal review policy has not yet been enforced; several PRs this iteration were merged by the author after CI passed. PRs still include links to related issues, and branch protection discourages direct commits to main. In the next sprint we plan to require at least one reviewer approval before merge and keep CI green as a hard gate.

### 3.3.4 Coding Standards

Code follows Google Java Style automatically via google-java-format; no manual style checks are needed. Public contracts are kept stable (for example, repository and security interfaces), and small, focused commits reference their task IDs using conventional prefixes (feat/fix/test/refactor/docs/chore). This keeps history readable and aligns implementation with user-story acceptance criteria.

#### app

| Element                                                                                                                      | Missed Instructions    | Cov. | Missed Branches        | Cov. | Missed | Cxty | Missed | Lines | Missed | Methods | Missed | Classes |
|------------------------------------------------------------------------------------------------------------------------------|------------------------|------|------------------------|------|--------|------|--------|-------|--------|---------|--------|---------|
|  <a href="#">org.soft2412.vsas.cli</a>      | <div><div></div></div> | 81%  | <div><div></div></div> | 58%  | 76     | 120  | 56     | 235   | 16     | 38      | 1      | 8       |
|  <a href="#">org.soft2412.vsas.repo</a>     | <div><div></div></div> | 87%  | <div><div></div></div> | 55%  | 49     | 82   | 28     | 150   | 0      | 22      | 0      | 2       |
|  <a href="#">org.soft2412.vsas.service</a>  | <div><div></div></div> | 71%  | <div><div></div></div> | 44%  | 16     | 22   | 13     | 43    | 1      | 5       | 0      | 1       |
|  <a href="#">org.soft2412.vsas.security</a> | <div><div></div></div> | 91%  | <div><div></div></div> | 80%  | 4      | 17   | 4      | 46    | 0      | 7       | 0      | 1       |
|  <a href="#">org.soft2412.vsas.model</a>    | <div><div></div></div> | 87%  | <div><div></div></div> | 100% | 1      | 19   | 1      | 35    | 1      | 18      | 0      | 2       |
|  <a href="#">org.soft2412.vsas</a>          | <div><div></div></div> | 25%  |                        | n/a  | 1      | 2    | 3      | 4     | 1      | 2       | 0      | 1       |
| Total                                                                                                                        | 403 of 2,412           | 83%  | 139 of 329             | 57%  | 147    | 262  | 105    | 513   | 19     | 92      | 1      | 15      |

#### Test Summary

|       |          |         |          |
|-------|----------|---------|----------|
| 32    | 0        | 0       | 0.081s   |
| tests | failures | ignored | duration |

100%  
successful

Packages

Classes

| Class                                                                | Tests | Failures | Ignored | Duration | Success rate |
|----------------------------------------------------------------------|-------|----------|---------|----------|--------------|
| <a href="#">org.soft2412.vsas.AppTest</a>                            | 1     | 0        | 0       | 0.014s   | 100%         |
| <a href="#">org.soft2412.vsas.FileScrollRepositoryTest</a>           | 3     | 0        | 0       | 0.014s   | 100%         |
| <a href="#">org.soft2412.vsas.ListCommandTest</a>                    | 2     | 0        | 0       | 0.009s   | 100%         |
| <a href="#">org.soft2412.vsas.SessionServiceTest</a>                 | 2     | 0        | 0       | 0.005s   | 100%         |
| <a href="#">org.soft2412.vsas.UploadCommandTest</a>                  | 4     | 0        | 0       | 0.013s   | 100%         |
| <a href="#">org.soft2412.vsas.WhoAmlCommandTest</a>                  | 2     | 0        | 0       | 0.004s   | 100%         |
| <a href="#">org.soft2412.vsas.cli.AuthHashingIntegrationTest</a>     | 1     | 0        | 0       | 0.006s   | 100%         |
| <a href="#">org.soft2412.vsas.cli.LoginCommandTest</a>               | 2     | 0        | 0       | 0.003s   | 100%         |
| <a href="#">org.soft2412.vsas.cli.RegisterCommandTest</a>            | 3     | 0        | 0       | 0.002s   | 100%         |
| <a href="#">org.soft2412.vsas.cli.RegisterCommandUniqueldKeyTest</a> | 1     | 0        | 0       | 0.001s   | 100%         |
| <a href="#">org.soft2412.vsas.model.UserTest</a>                     | 2     | 0        | 0       | 0.001s   | 100%         |
| <a href="#">org.soft2412.vsas.repo.FileUserRepositoryTest</a>        | 2     | 0        | 0       | 0.003s   | 100%         |
| <a href="#">org.soft2412.vsas.security.PasswordHasherTest</a>        | 7     | 0        | 0       | 0.006s   | 100%         |

```
app/src
├── main
│   ├── java
│   │   ├── org
│   │   │   ├── soft2412
│   │   │   │   ├── vsas
│   │   │   │   │   ├── App.java
│   │   │   │   │   ├── cli
│   │   │   │   │   │   ├── Command.java
│   │   │   │   │   │   ├── CommandDispatcher.java
│   │   │   │   │   │   ├── ListCommand.java
│   │   │   │   │   │   ├── LoginCommand.java
│   │   │   │   │   │   ├── LogoutCommand.java
│   │   │   │   │   │   ├── RegisterCommand.java
│   │   │   │   │   │   ├── UploadCommand.java
│   │   │   │   │   │   ├── WhoAmICommand.java
│   │   │   │   │   ├── model
│   │   │   │   │   │   ├── Scroll.java
│   │   │   │   │   │   ├── User.java
│   │   │   │   │   ├── repo
│   │   │   │   │   │   ├── FileScrollRepository.java
│   │   │   │   │   │   ├── FileUserRepository.java
│   │   │   │   │   │   ├── ScrollRepository.java
│   │   │   │   │   │   ├── UserRepository.java
│   │   │   │   │   ├── security
│   │   │   │   │   │   ├── PasswordHasher.java
│   │   │   │   │   ├── service
│   │   │   │   │   │   ├── SessionService.java
│   │   │   │   │   └── test
│   │   │   │   │       ├── java
│   │   │   │   │       │   ├── org
│   │   │   │   │       │   │   ├── soft2412
│   │   │   │   │       │   │   │   ├── vsas
│   │   │   │   │       │   │   │   │   ├── AppTest.java
│   │   │   │   │       │   │   │   │   ├── cli
│   │   │   │   │       │   │   │   │   │   ├── AuthHashingIntegrationTest.java
│   │   │   │   │       │   │   │   │   │   ├── LoginCommandTest.java
│   │   │   │   │       │   │   │   │   │   ├── RegisterCommandTest.java
│   │   │   │   │       │   │   │   │   │   ├── RegisterCommandUniqueIdKeyTest.java
│   │   │   │   │       │   │   │   │   ├── FileScrollRepositoryTest.java
│   │   │   │   │       │   │   │   │   ├── ListCommandTest.java
│   │   │   │   │       │   │   │   │   ├── model
│   │   │   │   │       │   │   │   │   │   ├── UserTest.java
│   │   │   │   │       │   │   │   │   ├── repo
│   │   │   │   │       │   │   │   │   │   ├── FileUserRepositoryTest.java
│   │   │   │   │       │   │   │   │   ├── security
│   │   │   │   │       │   │   │   │   │   ├── PasswordHasherTest.java
│   │   │   │   │       │   │   │   │   ├── SessionServiceTest.java
│   │   │   │   │       │   │   │   │   ├── UploadCommandTest.java
│   │   │   │   │       │   │   │   │   └── WhoAmICommandTest.java
```

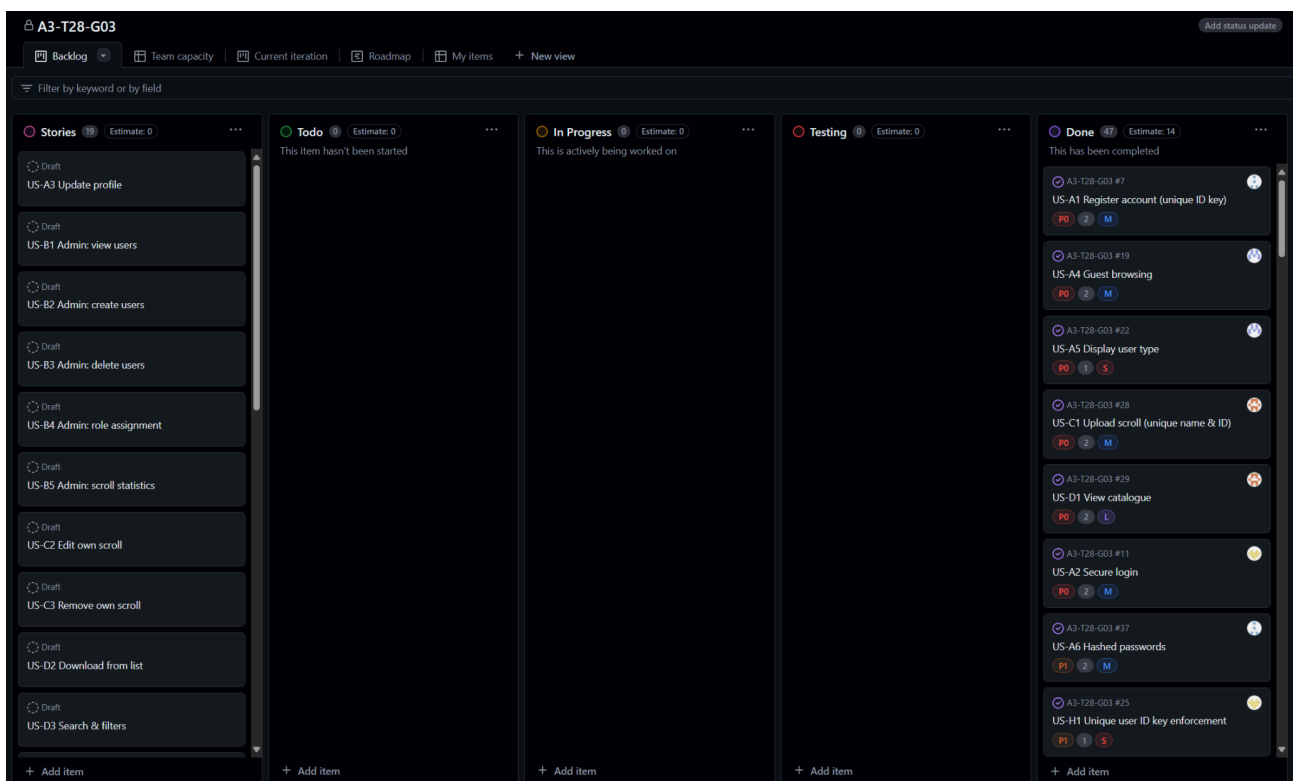
## 4. Application Development

### 4.1 Collaborative Coding

During Sprint 1, the team adopted a structured collaborative workflow on GitHub, emphasizing traceability, modularity, and disciplined version control.

Development followed an Epic–User Story–Task hierarchy. Each Epic represented a major functional domain (e.g., Account & Identity, Scroll Management), while User Stories were indexed using a consistent shorthand system (e.g., US-A1 for “Register account with unique ID key”).

This system enabled clear linkage between GitHub Issues, Pull Requests, commits, and Jenkins CI results, ensuring that every code change corresponded to a defined business objective.



Branching and commit conventions were standardised:

Branch names followed the pattern feature/register-account, clearly indicating the function.

|                                                |              |         |        |            |
|------------------------------------------------|--------------|---------|--------|------------|
| Appendix_A                                     | 16 hours ago | ✓ 1 / 1 | 2 / 1  | 🗑️ ...     |
| feat/scroll-list-command                       | 16 hours ago | ✗ 0 / 1 | 36 / 0 | 🗑️ ...     |
| feat/cli-implement-upload-command              | 16 hours ago | ✓ 1 / 1 | 34 / 0 | 🗑️ ...     |
| feat/US-A5-display-user-type                   | 16 hours ago | ✓ 1 / 1 | 24 / 0 | 🗑️ ...     |
| feature/cli-session                            | 16 hours ago | ✓ 1 / 1 | 13 / 0 | 🗑️ ...     |
| chore/gitignore-data                           | 16 hours ago | ✓ 1 / 1 | 11 / 0 | 🗑️ ...     |
| chore/format-google-java                       | 16 hours ago | ✓ 1 / 1 | 39 / 0 | 🗑️ ... #43 |
| feature/interactive-shell                      | 16 hours ago | ✓ 1 / 1 | 22 / 0 | 🗑️ ...     |
| feature/us-a1-Register-account                 | 16 hours ago | ✓ 1 / 1 | 20 / 0 | 🗑️ ...     |
| feature/us-a6-hashed-passwords                 | 16 hours ago | ✓ 1 / 1 | 38 / 0 | 🗑️ ...     |
| fix/cli-usage-help-text                        | 16 hours ago | ✓ 1 / 1 | 7 / 0  | 🗑️ ...     |
| fix/formatting-registercommand-authhashingtest | 16 hours ago | ✓ 1 / 1 | 27 / 0 | 🗑️ ...     |
| test/increase-coverage-rate                    | 16 hours ago | ✓ 1 / 1 | 2 / 0  | 🗑️ ...     |
| docs/readme-and-gitignore                      | 3 days ago   | ✗ 0 / 1 | 55 / 0 | 🗑️ ...     |
| chore/configure-project-tooling                | 3 days ago   | ✓ 1 / 1 | 45 / 0 | 🗑️ ...     |
| feature/sprint1-cli-skeleton                   | 3 days ago   | ✓ 1 / 1 | 40 / 0 | 🗑️ ...     |
| chore/bootstrap-with-gradle-init               | 3 days ago   | ✗ 0 / 1 | 53 / 0 | 🗑️ ...     |
| fix/ci-gradlew-and-format-check                | 3 days ago   | ✓ 1 / 1 | 42 / 0 | 🗑️ ...     |
| fix/ci-jenkinsfile-fix                         | 3 days ago   | ✓ 1 / 1 | 50 / 0 | 🗑️ ...     |
| feat/scroll/cli-upload-unique-id               | 3 days ago   | ✗ 0 / 1 | 38 / 0 | 🗑️ ...     |

Commit messages adhered to the Conventional Commits specification.

|                          |                                                                                                          |                                         |   |   |
|--------------------------|----------------------------------------------------------------------------------------------------------|-----------------------------------------|---|---|
| <input type="checkbox"/> | <b>set code style</b> ✓                                                                                  | #54 by xuyu8020 was merged 17 hours ago |   |   |
| <input type="checkbox"/> | <b>Test/increase coverage rate</b> ●                                                                     | #53 by xuyu8020 was merged yesterday    |   |   |
| <input type="checkbox"/> | <b>Fix/cli usage help text</b> ✓                                                                         | #52 by bzha8676 was merged 2 days ago   |   |   |
| <input type="checkbox"/> | <b>add "data" folder in gitignore</b> ✓                                                                  | #51 by xuyu8020 was merged 2 days ago   |   |   |
| <input type="checkbox"/> | <b>feat(cli): implement session persistence in Login and Logout commands</b> ✓                           | #50 by bzha8676 was merged 2 days ago   |   |   |
| <input type="checkbox"/> | <b>Feature/interactive shell</b> ✓                                                                       | #49 by bzha8676 was merged 2 days ago   |   |   |
| <input type="checkbox"/> | <b>US-A1: CLI registration with TSV storage and unique idKey (A3-T28-G03 #7)</b> ✓                       | #48 by xfan0282 was merged 2 days ago   | 7 |   |
| <input type="checkbox"/> | <b>feat(scroll): add list command with TSV-backed repositor</b> ✗                                        | #47 by zche0826 was merged 3 days ago   |   |   |
| <input type="checkbox"/> | <b>chore(format): re-run googleJavaFormat for RegisterCommand.java and A...</b> ✓                        | #46 by xfan0282 was merged 3 days ago   |   |   |
| <input type="checkbox"/> | <b>US-A6: Hashed passwords — implement salted SHA-256, register hashing, login verification, tests</b> ✓ | #45 by xfan0282 was merged 3 days ago   | 5 |   |
| <input type="checkbox"/> | <b>Feat/cli implement upload command and correct commit history</b> ✓                                    | #44 by zche0826 was merged 3 days ago   |   | 1 |

Each Pull Request required successful CI verification and peer review before merging.

This disciplined approach ensured that development progress was transparent and auditable. GitHub Projects visualised real-time progress through To Do / In Progress / Done columns, while automated Jenkins pipelines validated builds, tests, and code coverage. The workflow fostered collaborative accountability and allowed efficient parallel development across multiple functional areas.

## 4.2 Individual and Group Contributions

Each team member contributed to distinct functional modules while maintaining a unified development framework.

Boning Zhao was responsible for establishing the initial project scaffold, including the Gradle configuration, modular structure, and directory hierarchy.

This foundation ensured that all subsequent feature branches followed a consistent architecture and enabled smooth integration across modules.

Xuejian Fang handled the continuous integration setup, configuring the Jenkins multibranch pipeline and integrating Jacoco for automated coverage reporting.

His work ensured that every commit triggered automated build, testing, and coverage validation, laying the groundwork for reliable CI/CD practices throughout the project.

Following the infrastructure setup, each developer implemented their assigned feature modules and corresponding unit tests, ensuring traceable alignment with the defined User Stories:

Account and Identity (Epic A) – user registration (US-A1), secure login (US-A2), and password hashing (US-A6);

Scroll Management (Epic C) – upload functionality (US-C1) and validation mechanisms;

Browsing and Retrieval (Epic D) – basic catalogue view (US-D1) for guest users.

All members maintained task transparency using GitHub Issues.

This structure ensured consistent progress tracking, balanced workload distribution, and full traceability from requirements to implementation.

### 4.3 CI (Continuous Integration) Tools and Structure Management

The project adopted a continuous integration (CI) workflow centred on Jenkins and supported by Jacoco for automated test coverage tracking.

The CI configuration, implemented by Xuejian Fang, followed a multibranch pipeline design defined in the repository's Jenkinsfile.

Each commit or pull request automatically triggered a pipeline sequence consisting of Checkout → Build → Test → Coverage Report → Archive.

The Jenkins console output and archived HTML reports provided immediate feedback on build success, test results, and branch coverage, ensuring that integration issues were detected early.

The Jacoco plugin was integrated with the Gradle build process to measure coverage in every test execution.

Coverage thresholds were defined to enforce a minimum target of 75%, aligning with the project's non-functional requirements.

This automated process ensured that new commits maintained code quality and consistency, while preventing regressions in testing scope.

At the architectural level, the system was organised into modular Gradle subprojects, including CLI, Repository, Service, and Security layers.

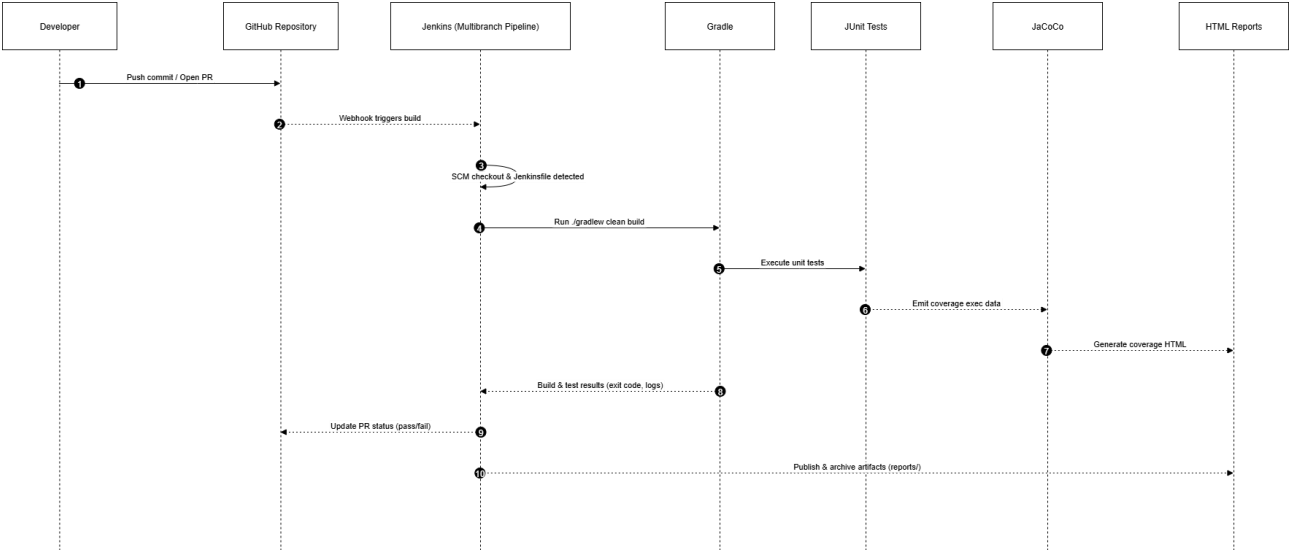
This separation of concerns allowed independent development and testing of each component, improving maintainability and reducing the risk of merge conflicts.

The modular build system also simplified CI configuration by enabling parallel execution of test tasks and consolidated coverage reporting across modules.

The combination of Jenkins automation and Gradle modularity established a reliable development environment that supported incremental delivery, reproducibility, and verifiable quality assurance across all project iterations.



Sequence Diagram



Class Diagram

```

class SessionService {
    SessionService(Path)
    SessionService()
    login() void
    login(User) boolean
    null isEmpty(String) String
    currentUser() Optional<User>
    trimToNull(String) String
    resolveDefaultSessionPath() Path
    parseInstant(String) Instant
}

```

```

class ScrollRepository {
    existsId(String) boolean
    findById(String) Optional<Scroll>
    save(Scroll) boolean
    findAll() List<Scroll>
}

```

```

class FileScrollRepository {
    FileScrollRepository()
    findById(String) Optional<Scroll>
    existsId(String) boolean
    findAll() List<Scroll>
    toTsv(Scroll) String
    nullToEmpty(String) String
    save(Scroll) boolean
    parseTsv(String) Optional<Scroll>
}

```

```

class UserRepository {
    existsIdKey(String) boolean
    findByUsername(String) Optional<User>
    findByIdKey(String) Optional<User>
    save(User) boolean
}

```

```

class User {
    User(String, String, String, String, String, String, String)
    User(String, String, String, String, String, String, String, String, String)
    email() String
    idKey() String
    role() String
    createdAt() Instant
    phone() String
    username() String
    salt() String
    passwordHash() String
    toString() String
}

```

```

class App {
    App()
    App(CommandInvoker)
    runInteractive(InputStream, PrintStream, PrintStream) int
    tokenize(String) String[]?
    main(String[]) void
    run(String[], InputStream, PrintStream, PrintStream) int
}

```

```

class FileUserRepository {
    FileUserRepository(Path)
    FileUserRepository()
    isBlank(String) boolean
    findByUsername(String) Optional<User>
    scanFirstMatch(String, String) Optional<User>
    rni(String) String
    indexOfCol(String[], String) int
    existsIdKey(String) boolean
    defaultRole(String) String
    save(User) boolean
    sanitize(String) String
    ensureHeader(Path) void
    findByIdKey(String) Optional<User>
}

```

```

class PasswordHasher {
    PasswordHasher()
    hexToBytes(String) byte[]
    bytesToHex(byte[]) String
    constantTimeEquals(byte[], byte[]) boolean
    hashToHex(char[], byte[]) String
    generateSalt(int) byte[]
    digest MessageDigest
}

```

```

class Scroll {
    Scroll(String, String, String, String, String, long)
    filePath() String
    id() String
    uploaderIdKey() String
    name() String
    downloadCount() long
    uploadDate() String
}

```

```

class Command {
    name() String
    run(String[]) int
    description() String
}

```

```

class CommandInvoker {
    dispatch(String[]) int
}

```

```

class ListCommand {
    ListCommand()
    name() String
    run(String[]) int
    description() String
}

```

```

class UploadCommand {
    UploadCommand()
    description() String
    name() String
    run(String[]) int
    parseOptions(String[]) Map<String, String>
}

```

```

class WhoAmICommand {
    WhoAmICommand()
    description() String
    name() String
    run(String[]) int
}

```

```

class LoginCommand {
    LoginCommand(PrintStream, PrintStream, PasswordHasher)
    LoginCommand()
    LoginCommand(PrintStream, PrintStream, PasswordHasher, SessionService)
    description() String
    findUserByUsername(Path, String) Optional<UserRow>
    run(String[]) int
    indexOfCol(String[], String) int
    name() String
}

```

```

class LogoutCommand {
    LogoutCommand()
    LogoutCommand(SessionService)
    run(String[]) int
    name() String
    description() String
}

```

```

class RegisterCommand {
    RegisterCommand()
    RegisterCommand(PrintStream, PrintStream, PasswordHasher, UserRow)
    RegisterCommand(PrintStream, PrintStream, PasswordHasher)
    run(String[]) int
    sanitize(String) String
    name() String
    description() String
}

```

```

class CommandDispatcher {
    CommandDispatcher()
    dispatch(String[]) int
}

```

```

class UserRow {
    UserRow(String, String, String, String)
    toUser(String) User
}

```

Jenkins

A3-T28-G03

Status

Configure

Scan Multibranch Pipeline Now

Scan Multibranch Pipeline Log

Multibranch Pipeline Events

Delete Multibranch Pipeline

Build History

Project Relationship

Check File Fingerprint

Favorite

Open Blue Ocean

Rename

Pipeline Syntax

Credentials

Build Queue

No builds in the queue.

Build Executor Status

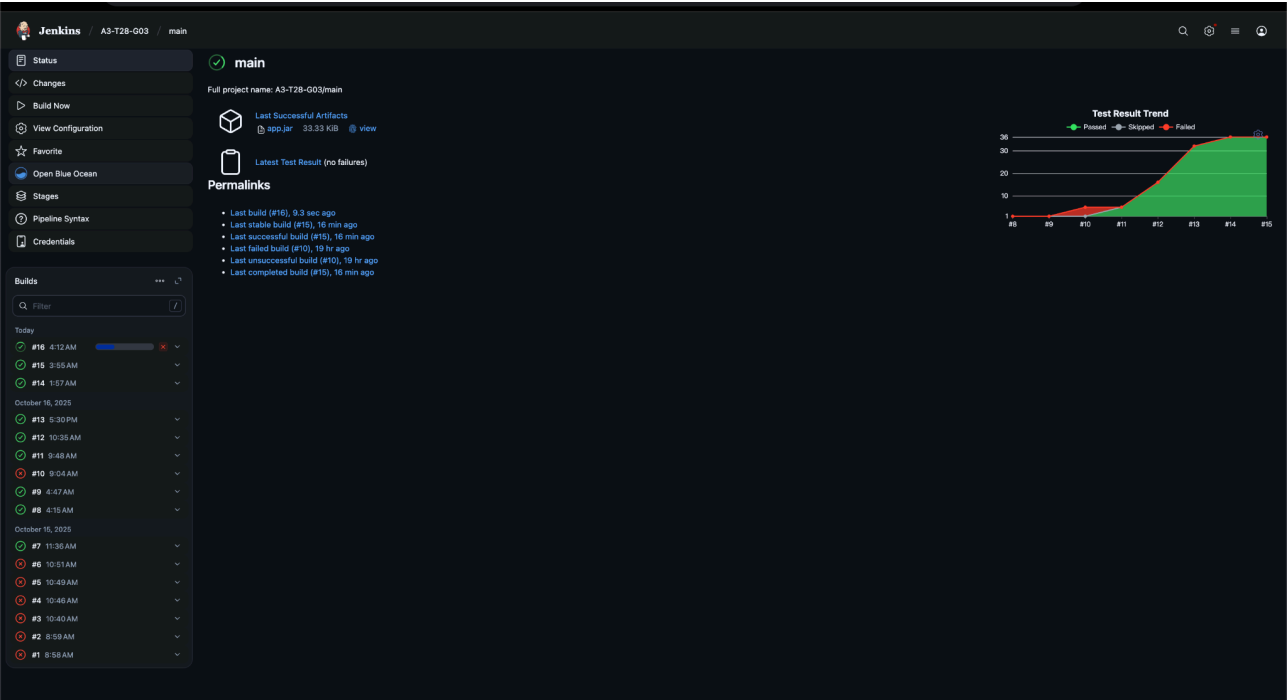
(0 of 2 executors busy)

A3-T28-G03

Branches (20)

| S | W  | Name ↓                                         | Last Success   | Last Failure   | Last Duration | F   |
|---|----|------------------------------------------------|----------------|----------------|---------------|-----|
| 🔴 | 🚧  | chore/bootstrap-with-gradle-init               | N/A            | 17 hr #4       | 2.2 sec       | ▶ ☆ |
| 🟢 | ☀️ | chore/configure-project-tooling                | 18 hr #2       | N/A            | 1 min 5 sec   | ▶ ☆ |
| 🟢 | ☀️ | chore/gitignore-data                           | 2 hr 13 min #1 | N/A            | 46 sec        | ▶ ☆ |
| 🔴 | 🚧  | docs/readme-and-gitignore                      | N/A            | 18 hr #2       | 1 min 33 sec  | ▶ ☆ |
| 🟢 | ☀️ | feat/cli-implement-upload-command              | 18 hr #1       | N/A            | 50 sec        | ▶ ☆ |
| 🔴 | 🚧  | feat/scrollcli-upload-unique-id                | N/A            | 17 hr #3       | 17 sec        | ▶ ☆ |
| 🔴 | 🚧  | feat/scroll-list-command                       | N/A            | 10 hr #1       | 1 min 7 sec   | ▶ ☆ |
| 🟢 | ☀️ | feat/US-A4-guest-browsing                      | 10 hr #1       | N/A            | 30 sec        | ▶ ☆ |
| 🟢 | ☀️ | feat/US-A5-display-user-type                   | 10 hr #1       | N/A            | 47 sec        | ▶ ☆ |
| 🔴 | 🚧  | feature/add-jenkinsfile                        | N/A            | 17 hr #2       | 2.7 sec       | ▶ ☆ |
| 🟢 | ☀️ | feature/cli-session                            | 2 hr 13 min #1 | N/A            | 1 min 2 sec   | ▶ ☆ |
| 🟢 | ☀️ | feature/interactive-shell                      | 2 hr 13 min #1 | N/A            | 26 sec        | ▶ ☆ |
| 🟢 | ☀️ | feature/sprint1-cli-skeleton                   | 18 hr #2       | N/A            | 47 sec        | ▶ ☆ |
| 🟢 | ☀️ | feature/us-a1-register-account                 | 10 hr #1       | N/A            | 27 sec        | ▶ ☆ |
| 🟢 | ☀️ | feature/us-a6-hashed-passwords                 | 10 hr #6       | N/A            | 1 min 26 sec  | ▶ ☆ |
| 🟢 | ☀️ | fix/ci-gradlew-and-format-check                | 18 hr #2       | N/A            | 1 min 8 sec   | ▶ ☆ |
| 🟢 | 🐛  | fix/ci-jenkinsfile-fix                         | 18 hr #3       | 1 day 16 hr #1 | 1 min 24 sec  | ▶ ☆ |
| 🟢 | ☀️ | fix/cli-usage-help-text                        | 2 hr 13 min #1 | N/A            | 26 sec        | ▶ ☆ |
| 🟢 | ☀️ | fix/formatting-registercommand-authhashingtest | 10 hr #1       | N/A            | 49 sec        | ▶ ☆ |
| 🟢 | ☀️ | main                                           | 15 min #15     | 19 hr #10      | 22 sec        | ▶ ☆ |

localhost:8080/job/A3-T28-G03/



```

[Pipeline] sh
15:12:04 + ./gradlew verifyGoogleJavaFormat --no-daemon
15:12:04 To honour the JVM settings for this build a single-use Daemon process will be forked. For more on this, please refer to https://docs.gradle.org/8.7/userguide/gradle_daemon.html#sec:disabling_the_daemon in the
Gradle documentation.
15:12:05 Daemon will be stopped at the end of the build
15:12:08 > Task :app:verifyGoogleJavaFormat
15:12:08
15:12:08 BUILD SUCCESSFUL in 4s
15:12:08 1 actionable task: 1 executed
[Pipeline] echo
15:12:09 ✓ Code format check passed.
[Pipeline] }
[Pipeline] // script
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Build)
[Pipeline] echo

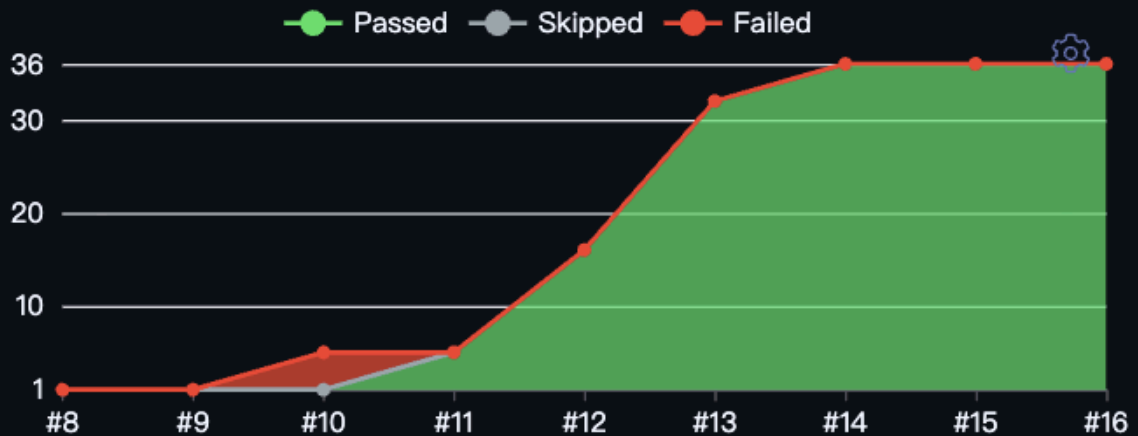
```

```

[Pipeline] echo
15:12:21 Pipeline completed successfully.
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // timestamps
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS

```

## Test Result Trend



Container CPU usage ⓘ

0.55% / 1000% (10 CPUs available)

Container memory usage ⓘ

839.1MB / 7.47GB

Show charts

Q Search



Only show running containers

| <input type="checkbox"/> | Name            | Container ID | Image       | Port(s)                                        | CPU (%) | Last s | Actions |
|--------------------------|-----------------|--------------|-------------|------------------------------------------------|---------|--------|---------|
| <input type="checkbox"/> | jenkins-blueoce | 07f5a47c3ac8 | myjenkins-l | 50000:50000 <a href="#">Show all ports (2)</a> | 0.55%   | 2 hour |         |

## 4.4 Functional and Non-functional Aspects

During Sprint 1, the project delivered a Minimum Viable Product (MVP) that met the essential functional and non-functional objectives.

## Functional Aspects

Functionally, the system implemented the core user operations necessary for a working prototype:

Account Registration and Authentication – users can register with a unique ID key (US-A1) and securely log in using hashed credentials (US-A2, US-A6);

Scroll Management – authenticated users can upload scrolls with unique identifiers (US-C1), while guests are allowed to browse available content in a read-only catalogue view (US-D1);

Data Validation and Access Control – the system enforces unique constraints (US-H1) to prevent duplicate accounts and ensure secure access isolation between user sessions.

The integration of these features created a fully operational command-line interface that supports user identity management and basic content handling, representing a complete vertical slice of the intended system.

## Non-functional Aspects

Non-functionally, the system satisfied key engineering goals defined in Epic F (Non-functional & Delivery):

Security – implemented salted password hashing to protect stored credentials and restricted unauthorised operations through access control checks;

Reliability and Maintainability – modular architecture under Gradle improved code separation and testing isolation;

Automation and Quality Assurance – integrated Jenkins and Jacoco ensured continuous testing and maintained coverage above the 75% threshold;

Together, these aspects established a stable and testable foundation for subsequent sprints, ensuring that future iterations can focus on expanding functionality without compromising software quality.

```
vsas> whoami
guest
vsas> upload --id 1 --name readme --file "C:\Users\32234\code\Java\2412-2025\A3-T28-
G03\README.md"
vsas> upload: login required
```

```
vsas> register --username test1 --password 00000000 --email 123@321.com --phone 123123123 --id-key 1
Registered user test1
vsas> register --username test2 --password 00000000 --email 123@321.com --phone 123123123 --id-key 2
Registered user test2
```

| username | email       | phone     | idKey | role | passwordHash                                                      | salt                             | createdAt                      |
|----------|-------------|-----------|-------|------|-------------------------------------------------------------------|----------------------------------|--------------------------------|
| test1    | 123@321.com | 123123123 | 1     | USER | 454af3bc588f75602ad7b04dfb60d3aecdc35a627cb948cdfb360fc9775a9c82a | 7bbce533ef16bfd73d69da7f4cf02a55 | 2025-10-19T05:31:31.280205700Z |
| test2    | 123@321.com | 123123123 | 2     | USER | 1deddf34a33da4b0f47ecb3195fed4c6c8894322dae4da8f963203f4466bc03   | 33656a0d00b9d3933da23dd471819e6b | 2025-10-19T05:31:47.196167200Z |

```
vsas> login --username test1 --password 00000000
Login success
vsas> whoami
test1 (role=USER)
vsas> upload --id 1 --name readme --file "C:\Users\32234\code\Java\2412-2025\A3-T28-G03\README.md"
upload: success
vsas> list
id | name | uploader | uploadDate
1 | readme | 1 | 2025-10-19T05:33:52.850998200Z
vsas> logout
Logout success
vsas> whoami
guest
vsas> list
id | name | uploader | uploadDate
1 | readme | 1 | 2025-10-19T05:33:52.850998200Z
```

## 5. Group Contribution

### 5.1 Team Organisation and Roles

The team followed the Scrum methodology to maintain a self-organised and iterative development process.

Roles were clearly defined in accordance with the Scrum structure:

Product Owner – Boning Zhao

Responsible for defining user requirements, prioritising backlog items, and ensuring that implemented features aligned with stakeholder expectations.

In addition to managing the product backlog, he contributed to the system's architecture by creating the initial Gradle-based scaffold and standardising the project's directory hierarchy.

Scrum Master – Xuejian Fang

Oversaw sprint planning, progress tracking, and facilitated team communication to ensure adherence to Scrum principles.

From a technical perspective, he implemented the CI/CD infrastructure, configuring the Jenkins multibranch pipeline and integrating Jacoco for automated coverage validation.

Core Team Members – Other Developers

Implemented assigned functional modules according to their respective Epics and User Stories, including user registration, authentication, and scroll management features. Each member maintained testing accountability for their code.

This role allocation balanced project management responsibilities and technical development, enabling the team to maintain both delivery speed and software quality throughout the sprint.

## 5.2 Communication and Collaboration Tools

The team primarily relied on in-person meetings as the main mode of communication during the project.

Regular offline discussions were held after class or in the university workspace, allowing members to clarify requirements, review progress, and resolve implementation issues efficiently.

Face-to-face collaboration proved particularly effective for debugging and rapid decision-making, as it enabled real-time code review and immediate feedback.

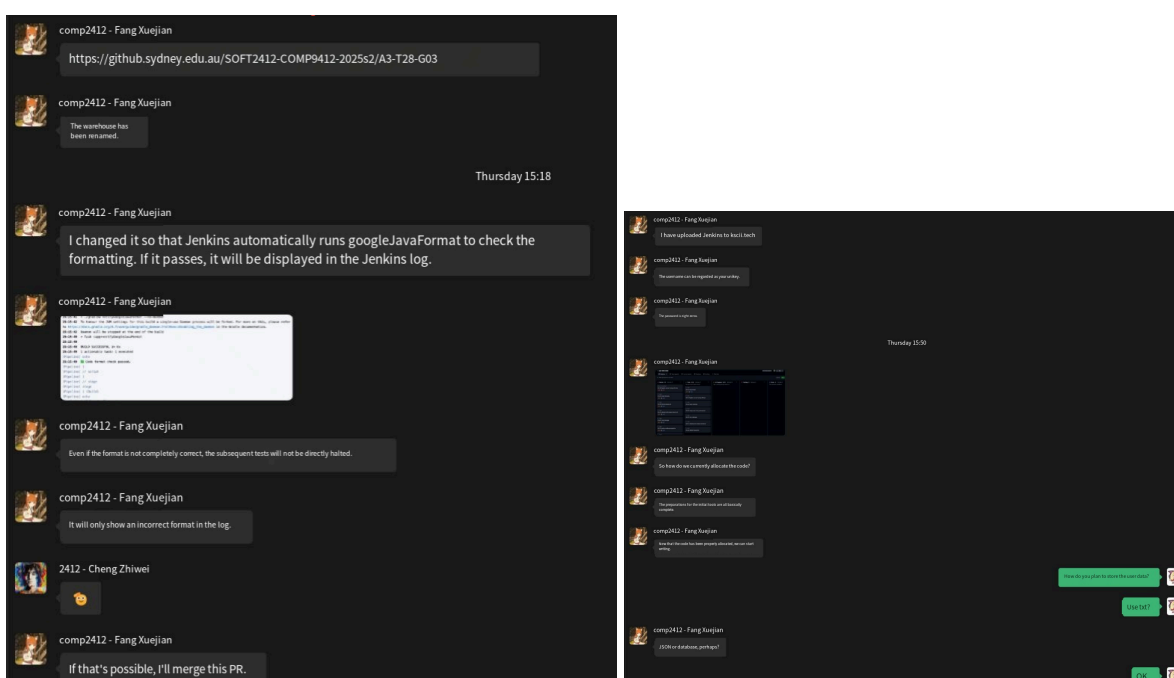
Outside of scheduled meetings, the team maintained continuous contact through WeChat, which served as a supplementary communication platform.

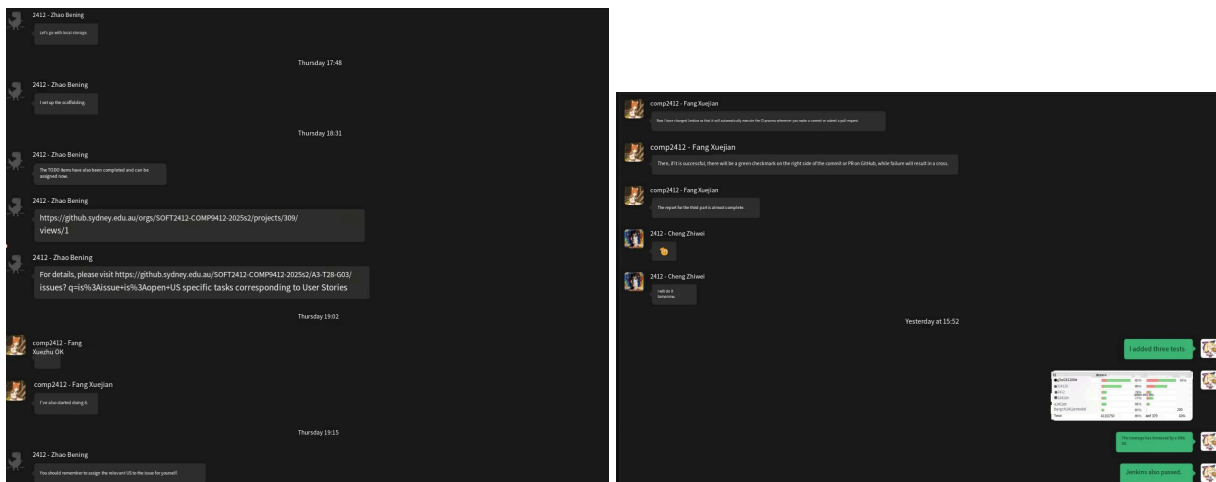
WeChat group chats were used to share updates, coordinate meeting times, and exchange files such as screenshots or build reports.

This hybrid approach—combining offline discussions with instant online messaging—ensured that communication remained both efficient and flexible, even when team members had different schedules.

To support task tracking and version control, GitHub was used for managing issues, reviewing pull requests, and maintaining code consistency.

Together, these tools created a collaborative workflow that balanced face-to-face engagement with digital coordination.





## 5.3 Group Discussion

Throughout the sprint, the team held regular in-person meetings to track progress, refine requirements, and resolve implementation challenges.

Most meetings took place on campus after class, where members collaboratively reviewed task status, shared technical issues, and aligned feature development.

This face-to-face communication proved efficient for debugging, rapid decision-making, and ensuring consistency across modules.

For asynchronous communication, WeChat was used as the supplementary tool to coordinate meeting times, share screenshots, and exchange updates between sessions.

This hybrid communication model maintained flexibility and ensured continuous progress even when members worked independently.

At the end of Sprint 1, the team conducted a demo presentation showcasing the working Minimum Viable Product (MVP).

Xuanhao Yu and Zhiwei Cheng demonstrated the functional aspects of the application, including user registration, login, scroll upload, and catalogue browsing, while verifying that each operation executed correctly through the command-line interface.

Meanwhile, Boning Zhao and Xuejian Fang presented the testing coverage results and Jacoco reports, highlighting that all test cases passed successfully and the overall branch coverage reached approximately 83%.

The demonstration provided a comprehensive overview of both the system's core functionality and its technical quality assurance.

The tutor's feedback was generally positive, commending the team's modular design, clear repository organisation, and effective CI pipeline integration.

However, it was also noted that the user interface (UI) lacked user-friendliness and could be improved in future iterations to enhance usability and visual clarity.



The team has acknowledged this feedback and plans to address it in the next sprint.

## 5.4 Individual Contributions

Each team member made distinct technical and non-technical contributions during Sprint 1. The responsibilities were divided according to the project's Epics and user stories to ensure balanced workload and clear accountability.

### Boning Zhao (Product Owner)

Technical contributions: Defined the overall project scope and initial user stories based on client requirements. Set up the Gradle project scaffold, configured the build system, and implemented key stories — US-A2 Secure Login and US-H1 Unique User ID Key Enforcement — establishing the foundation for authentication and identity management.

Non-technical contributions: Prioritised backlog items, clarified acceptance criteria, and ensured alignment between client expectations and team implementation.

### Xuejian Fang (Scrum Master)

Technical contributions: Implemented US-A1 Register Account (unique ID key) and US-A6 Hashed Passwords, ensuring secure credential handling and unique ID validation. Configured and deployed the Jenkins CI pipeline and Jacoco coverage reports, achieving approximately 85% test coverage.

Non-technical contributions: Organised Scrum meetings, maintained sprint documentation, and ensured the team adhered to Agile principles.

### Zhiwei Cheng (Core Developer)

Technical contributions: Implemented US-C1 Upload Scroll (unique name & ID) and US-D1 View Catalogue, building the upload logic, file metadata retrieval, and catalogue display. Improved project structure and documentation for better maintainability.

Non-technical contributions: Assisted in integration testing and co-presented the functional demo showing file upload and viewing features.

### Xuanhao Yu (Core Developer)

Technical contributions: Developed US-A4 Guest Browsing and US-A5 Display User Type, enabling non-logged-in users to view scrolls and display user roles dynamically.

Non-technical contributions: Participated in sprint discussions, contributed to peer code review, and demonstrated the system's runtime functionality during the final presentation.

Collectively, the team delivered a stable and functional Minimum Viable Product (MVP) with 85% Jacoco test coverage, automated build and test workflows, and a maintainable modular codebase.

## 5.5 Additional Notes and Reflections

During Sprint 1, the team reflected on its performance across development, collaboration, and presentation activities.

Overall, the sprint was successful in achieving its planned goals — delivering a functional MVP with automated testing and CI/CD integration — but several lessons and areas for improvement were identified.

First, the sprint reinforced the importance of clear task ownership and early integration.

While the team successfully divided work by user stories, occasional delays occurred when merging branches with overlapping code.

To address this, the group plans to hold more frequent code review sessions and maintain a shared integration branch to detect issues earlier.

Second, the demo presentation provided valuable feedback from the tutor, who commended the team's structured workflow, CI setup, and testing discipline but noted that the user interface (UI) lacked user-friendliness.

Although the current command-line prototype functioned correctly, its usability and clarity can be improved.

Therefore, the next sprint will prioritise UI design refinement, focusing on clearer layouts, improved interaction feedback, and better visual consistency.

Overall, Sprint 1 established a solid foundation for both the system architecture and the team's workflow.

The key takeaways include stronger emphasis on integration testing, better time allocation for UI improvements, and continued focus on collaboration quality to ensure consistent delivery in future sprints.